MILITARY SPECIFICATION

BALLISTICS COMPUTER SUBASSEMBLY: 8635412 (CANT UNIT)

1. SCOPE

1.1 This specification covers one type of electrical mechanical assembly known as Ballistics Computer Subassembly: 8635412, hereinafter referenced as "Cant Unit" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

MIL-P-14232 —Parts Equipment and Tools for Ordnance Materiel: Packaging of

MIL-I-45208 —Inspection System Requirements

MIL-I-45607 —Inspection Equipment Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 —Quality Assurance Terms and Definitions

MIL-STD-202 —Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

F8635412 —Ballistics Computer Subassembly: F8635412 (Cant Unit)

IEL 8635412—List of Inspection Documents for Ballistics Computer Subassembly: 8635412 (Cant Unit)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials should be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. The cant unit shall be manufactured in accordance with Drawing F8635412 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.3 General specification. The contractor

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shall be responsible for adherence to and compliance with the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 *Operating temperatures.* The cant unit shall meet all the requirements of 3.6 while exposed and thermally stabilized at ambient temperatures of −40° and +150° Fahrenheit (F). Upon return to standard ambient temperature the cant unit shall meet the requirements of 3.4.2.

3.4.2 *Vibration.* The cant unit shall meet the requirements of 3.5 to 3.6.3 inclusive after being vibrated in a vertical plane for five minutes plus or minus 15 seconds at an amplitude of 1/16 inch (⅛ inch total excursion) at a frequency of 30 cycles per second.

3.4.3 *Water resistance (sealing).* The interior of the cant unit shall show no evidence of water penetration or other damage after being subjected to a simulated rainfall for a time duration of 15 minutes ± 15 seconds.

3.5 Electrical.

3.5.1 *Continuity.* The continuity of the cant unit shall conform to Electrical Schematic Drawing C10516514.

3.5.2 *Dielectric strength.* There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the cant unit are subjected to an alternating current (ac) root mean square (rms) potential of 300 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of not more than 30 seconds nor less than five seconds.

3.5.3 *Insulation resistance.* All insulation between ungrounded circuits and ground and between isolated circuits, shall have a minimum resistance of 10 megohms with 500 volts dc applied.

3.6 Performance.

3.6.1 *Resolver input travel.* The magnet arm shall contact the stops limiting travel at cant unit housing rotations of 15 degree, minus zero plus 3.0 minutes, clockwise and counterclockwise from a base level reference.

3.6.2 *Magnet arm damping.* The magnet arm shall return to its normal gravity controlled position with a maximum of one overshoot when displaced at least 5 degrees.

3.6.3 *Output voltages.* The cant unit output voltage ratios shall conform to the requirements of Table I in Drawing F8635412, Sheet 2, when the unit is energized in conformance with the test circuit shown thereon. The minimum null indication, neglecting test circuit switching spikes, shall not exceed 4.0 millivolts for cosine readings and 1.0 millivolt for sine readings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or purchase order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of MIL-I-45208 and the component inspection requirements

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of MIL–F–13926. Definitions of inspection terms not otherwise defined herein shall be as listed in MIL–STD–109.

4.2 **Inspection provisions.**

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL–STD–105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL–F–13926 and the Supplementary Quality Assurance Provisions, listed on IEL 8635412. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL–STD–105, and the inspection level and sampling plan specified in table I titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL–P–14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE I. *Classification of Defects*

Use Level II of Table I with
Sampling Plan Table II-A of MIL–STD–105

CRITICAL: None
MAJOR: AQL 1.0%

	Requirement	Test Procedure
101. Physical failure, post vibration	3.4.2	4.5.1.2
102. Continuity	3.5.1	4.5.2
103. Resolver input travel	3.6.1	4.5.5.1
104. Magnet arm dampening	3.6.2	4.5.5.2
105. Output voltages	3.6.3	4.5.5.3

MINOR: None

Note. The above tests shall be performed at standard ambient temperature (60° to 90° F.).

4.2.3 *Disposition of non-conforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL–STD–105.

4.3 **Control tests.**

4.3.1 *General.* One cant unit shall be selected at random from each one hundred produced, or from each month's production, whichever occurs first. The control sample shall meet the following requirements and tests:

Control test	Requirement	Test Procedure
301. Materials	3.1	as per spec. or dwg.
302. Fabrication	3.2	visual—per dwg.
303. General specification	3.3	visual—MIL–F–13926

4.3.2 *Environmental and electrical.* Three cant units shall be selected at random from each fifty produced, or from each month's production, whichever occurs first. The control sample shall meet the following requirements and tests:

Control test	Requirement	Test Procedure
304. Operating temperature	3.4.1	4.5.1.1
305. Water resistance	3.4.3	4.5.1.3
306. Dielectric strength	3.5.2	4.5.3
307. Insulation resistance	3.5.3	4.5.4

4.3.3 *Control test failure.* Should any one item of the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 **Inspection equipment.** Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirement of MIL–I–45208 and all applicable requirements specified in MIL–I–45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

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4.4.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of MIL-F-13926. Concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2 *Inspection equipment accuracy requirements.* The devices required in the test methods and procedures herein, to measure the electrical and mechanical values specified in Section 3 shall provide readings to the accuracy specified below:

Item Number	*Description*
1. Ratio standard	Ratio range from .00000 to 1.00000 accuracy to ± .01%. Input impedance at 400 cps, 10,000 ohms minimum. Series impedance—10 ohms maximum. Phase shift 0.01° maximum.
2. Oscilloscope	Minimum sensitivty of 20 millivolts, full scale, vertical channel. External synchronization. Input impedance—1 megohm ± 10%.
3. A.C. power supply	0–45 volts rms, 400 cps ± 2 cps, 0.5% maximum harmonic distortion, maximum variation in distortion during test 0.05%.
4. Testing fixture	Shall mount the cant unit in a true horizontal position. Rotate the cant unit up to 20° in either direction about the resolver axis with settings to 5°, 10° and 15° within 2.0 minutes. Provision for instantaneous release from a 20° position and return to horizontal position.

4.5 Test methods and procedures.

4.5.1 *Environmental.*

4.5.1.1 *Operating temperatures.* This test shall be conducted in conformance with the temperatures specified in 3.4.1. The test equipment shall be in accordance with the "Test Facilities" requirements specified in MIL-F-13926. While thermally stabilized at the specified temperatures, the cant unit shall be subjected to the tests specified in 4.5.5 to 4.5.5.3 inclusive to determine conformance with the requirements referenced therein.

4.5.1.2 *Vibration.* To determine conformance to 3.4.2, the cant unit shall be mounted in its normal position using a suitable mounting apparatus to attach it to a vibrating machine or fixture. Test equipment shall be in accordance with the "Test Facilities" portion of MIL-F-13926 and shall be capable of providing vibration at the amplitude and frequency for the time specified in 3.4.2. After vibration, the cant unit shall be given a visual-tactile examination for damage and loose parts. The cant unit shall then be subjected to the tests specified in 4.5.2 to 4.5.5.3 inclusive to determine conformance with the respective requirements referenced therein.

4.5.1.3 *Water resistance.* To determine conformance to 3.4.3, this test shall be conducted in accordance with Procedure I of MIL-F-13926 except for the time duration specified in 3.4.3. The cant unit shall show no evidence of water penetration as a result of this test when visually examined.

4.5.2 *Continuity.* The cant unit shall be tested, using standard measuring equipment, to demonstrate continuity of the circuits in accordance with Schematic Drawing D8635307 to determine compliance with 3.5.1.

4.5.3 *Dielectric strength.* The cant unit shall be tested in accordance with Method 301 of MIL-STD-202 and shall meet the requirements specified in 3.5.2.

4.5.4 *Insulation resistance.* Immediately after satisfactorily withstanding the test for dielectric strength, the cant unit shall be given the insulation resistance test in accordance with Method 302, test condition B, of MIL-STD-202, and shall meet the requirements specified in 3.5.3.

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4.5.5 *Performance tests.* These tests shall be performed using the applicable testing devices specified in 4.4.2, and in accordance with the test circuit of Drawing F8635412.

4.5.5.1 *Resolver input travel.* To determine conformance to 3.6.1 rotate the housing on the testing fixture in a clockwise direction against its stop and observe the angle at which the output voltage hits a constant value. At that point read the angle of rotation. Repeat the procedure in the opposite direction.

4.5.5.2 *Magnet arm damping.* To determine conformance to 3.6.2, physically rotate the housing 20 degrees about the centerline of the resolver axis. At this point, an instantaneous release and return of the housing to the original horizontal position of the locating surfaces shall cause the magnet arm to return to its normal gravity controlled position within one overshoot as specified in 3.6.2. One overshoot shall be determined by observing the oscilloscope null on the test circuit (see 4.4.2).

4.5.5.3 *Output voltages.* To determine conformance to 3.6.3 mount the cant unit to the testing device specified in 4.4.2 and connect the unit to the test circuit specified in 4.5.5. With the proper input voltage and ratio standard setting, check the output on the oscilloscope (4.4.2) for square wave display. Set the cant unit to 0 degrees and position S1 and S2 switches as shown on table I of Drawing F8635412. Observe ratio setting for conformance to the tolerance specified. Continue to test for all right and left cant angles listed in the referenced table I.

5. PREPARATION FOR DELIVERY

5.1 *Packaging, packing and marking.* Packaging, packing and marking shall be in accordance with MIL-P-14232 and applicable packaging data sheet. The level of protection shall be as specified in the procurement document (see 6.2).

6. NOTES

6.1 Intended use. The cant unit is primarily intended as a functional component of the electronic Computer, Ballistics: M16. The cant unit is designed to measure gun trunnion cant angles, convert these angles into electrical voltage functions of sine and cosine and supply the functions to the computer system where they are used to multiply ballistics correction factors when the weapon is sighted on target with the gun trunnion canted.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of protection.
(c) Applicable packaging data sheets (see 5.1).
(d) Applicable stock number.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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